

Module 7: Photosynthesis & Cellular Respiration —

Study Questions

Photosynthesis

1. Write the balanced equation for photosynthesis. Label the reactants and products.
2. Where does photosynthesis occur? Describe the structure of a chloroplast (thylakoid, granum, stroma).
3. What is chlorophyll? Why do most plants appear green?
4. Where do the light-dependent reactions occur? What are their inputs and outputs?
5. Trace the path of an electron from water through both photosystems to NADPH.
6. Why is water split during the light reactions? What gas is released as a byproduct?
7. Explain chemiosmosis in the thylakoid membrane. How does the H^+ gradient drive ATP synthesis?
8. Where does the Calvin Cycle occur? What are its inputs and outputs?
9. What is carbon fixation? What enzyme catalyzes it?
10. How many turns of the Calvin Cycle are needed to produce one molecule of glucose?
11. Compare C_3 , C_4 , and CAM plants. How do C_4 and CAM plants minimize photorespiration?

Cellular Respiration

1. Write the balanced equation for cellular respiration. How does it relate to the photosynthesis equation?
2. Where in the cell does glycolysis occur? Does it require oxygen?

3. What are the inputs and outputs of glycolysis (per glucose molecule)?
4. What happens during pyruvate oxidation? Where does it occur?
5. Describe the Krebs Cycle. What are its inputs and outputs (per glucose)?
6. By the time the Krebs Cycle is complete, what has happened to all six carbons from the original glucose?
7. What is the Electron Transport Chain? Where is it located?
8. Explain oxidative phosphorylation. How does the flow of electrons lead to ATP production?
9. What is the role of oxygen in cellular respiration? Why is it called the "final electron acceptor"?
10. What is the total ATP yield from one glucose molecule (aerobic respiration)?
11. Why is the ATP yield given as a range (30-32) rather than a fixed number?

Fermentation

1. What is fermentation? When does a cell use fermentation instead of aerobic respiration?
2. Compare lactic acid fermentation and alcohol fermentation. Where does each occur?
3. Does fermentation produce new ATP? What is its primary purpose?

Connecting the Processes

1. Complete this statement: "Photosynthesis is to _ **as cellular respiration is to** ____." (anabolic/catabolic)
2. Both photosynthesis and cellular respiration use chemiosmosis. Compare where it occurs and what drives the H^+ gradient in each process.

3. Explain how matter (carbon, oxygen, hydrogen) cycles between photosynthesis and cellular respiration in the biosphere.
4. If a plant is placed in the dark for an extended period, what happens to its glucose supply? Which process can it still perform?
5. Proteins and fats can also enter the respiratory pathway. Where do fatty acids enter? Where do amino acids enter after deamination?